



Feature importance in the age of explainable AI: Case study of detecting fake news & misinformation via a multi-modal framework

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ABSTRACT

In recent years, fake news has become a global phenomenon due to its explosive growth and ability to leverage multimedia content to manipulate user opinions. Fake news is created by manipulating images, text, audio, and videos, particularly on social media, and the proliferation of such disinformation can trigger detrimental societal effects. False forwarded messages can have a devastating impact on society, spreading propaganda, inciting violence, manipulating public opinion, and even influencing elections. A major shortcoming of existing fake news detection methods is their inability to simultaneously learn and extract features from two modalities and train models with shared representations of multimodal (textual and visual) information. Feature engineering is a critical task in the fake news detection model's machine learning (ML) development process. For ML models to be explainable and trusted, feature engineering should describe how many features used in the ML models contribute to making more accurate predictions. Feature engineering, which plays an important role in the development of an explainable AI system by shaping the features used in the ML models, is an interconnected concept with explainable AI as it affects the model's interpretability. In the research, we develop a fake news detector model in which we (1) identify several textual and visual features that are associated with fake or credible news; specifically, we extract features from article titles, contents, and, top images; (2) investigate the role of all multimodal features (content, emotions and manipulation-based) and combine the cumulative effects using the feature engineering that represent the behavior of fake news propagators; and (3) develop a model to detect disinformation on benchmark multimodal datasets consisting of text and images. We conduct experiments on several real-world multimodal fake news datasets, and our results show that on average, our model outperforms existing single-modality methods by large margins that do not use any feature optimization techniques.

1. Introduction

The recent transformation of social media channels (Facebook, Twitter, YouTube, and WhatsApp, etc.) has changed the whole way we lead our lives and acquire information. Prior to approximately a decade ago, our primary source of news was traditional journalism, established media organizations, and other reliable sources that adhered to specific ethical standards (Hunt & Matthew, 2017). Most of the news we read today, particularly via social media feeds and tweets, might appear true to our eyes; however, that often is not the case. The intent behind misinformation spread is to deceive the readers, and the extensive growth of disinformation spread on media channels has particularly exploded in the last few years via platforms such as YouTube, LinkedIn, Facebook, Instagram, WhatsApp, Twitter, and Sina Weibo, among

others. Whether through conspiracy planning or pandemic dealing, social media sites have transformed the dissemination mode of information, and these platforms have always functioned as a platform for 'fake news' (Zafarani et al., 2019; Agarwal et al., 2022; Zhang et al., 2019). The extensive use of social websites to disseminate false and often incendiary information stirred alarm and highlighted the detection of fake news as a critical issue that needed serious attention.

Ironically, the term 'fake news' was actually introduced by Donald Trump to highlight his accusations against the opposing party during the aforementioned US elections; however, he arguably emerged as the greatest beneficiary of the proliferation of misinformation. For example, a widespread claim that Trump fed police officers working protests in Chicago was initiated by a single tweet from a person who was not even present in the city. A study conducted during the election year found

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that 88 % of US citizens believed that misunderstandings were being fomented by false news; however, 25 % of survey participants accepted the sharing of fake political news online due to ignorance (Barthel, Mitchell, & Holcomb, 2016). Another instance was also related to the 2016 US presidential elections, which was immensely driven by misleading information on social sites. On Twitter, nearly 529 different rumours were spread, and malicious bot accounts which counted for an average of 19 million, tweeted or retweeted tweets in favor of Trump or Clinton. According to research (Silverman, 2016), fake stories and misinformation were more widely used on Twitter, WhatsApp, and Facebook platforms than the real stories in the 2016 election. Silverman and Singer-Vine (2016) proved that most of the people who read false news and misinformation or watch fake videos or stories on Facebook and Twitter, believe them and they also proved that most discussed fake news stories in the 2016 USA presidential election tended to favor Donald Trump over Hillary Clinton (Papanastasiou, 2020; Chang et al., 2022). A recent study conducted in the USA tells that (i) 60 % of people in the United States received the news online from an online media platform (Gottfried & Shearer, 2016) (ii) Most of the time fake news and misinformation was shared on Facebook and Twitter than any other platforms on social media (Silverman, 2016) (iii) Most people report to believe the news to be true which is actually fake news (Silverman and Singer-Vine, 2016). Ever since these cases, various departments including social media companies came under pressure to build something against the expansion of misinformation on media channels.

The widespread disinformation can negatively affect people's behavior and cause the detrimental societal effects. The Impact of fake forwarded messages is well known and how it is plotted to disturb societal peace with the intent to spread violence, cause riots, humiliate emotions, influence elections, or initiate propaganda (Kim & Dennis, 2020; Moravec et al., 2020; Reisach, 2021; Tanınmış et al., 2022). Another example is about Las Vegas massacre in October 2017, which reportedly killed 59 people and injured more than 500 people, began spreading on Facebook and Google with false reports of the deadly tragedy. In 2018, a theme issue about 'Fake News', which reported that fake stories tend to arise human feelings of fear and surprise (Vosoughi et al., 2018) leading to social panic, was published by Science magazine. An example of disinformation impact is a fake video with the name Somalis 'pushed into shallow grave' Ethiopia that lead to a fierce clash between two races in Ethiopia. Another clash between Greek police with travelers or migrants was caused due to rumor spread on social media stating the elevation of onward travel restrictions in Greece. These examples explain the serious threat of the increasing spread of fake news in society (David et al., 2018).

Fake news, misinformation, or disinformation are generally created and shared on social media by manipulating or tempering videos, text, and images. Although technology advancements and enhanced AI tracking have shown great progress in the detection of anomalous information or fake news, but the threat still persists because of the complexity created by disinformation's volume and velocity (George et al., 2018; Han et al., 2020). The issues created by fake news spread through various social media sites and other popular communication apps are worth taking note of for many reasons. There is a lot that goes behind the misinformation spreading such as alteration, distortion, or manipulation of texts, images, and videos with the help of several other contents. There are software and online platforms available with multiple features to manipulate the content and publish it as pieces of real news. Fake news makes the best possible use of videos and images to give an intuitive experience for readers. Facebook, Twitter, and other online media channels have transformed the whole news market evolving from text-only news to news with multimodal news which has images, text, audio, and videos. This transformation provides an interactive storytelling experience and has the magical power to delight and engage readers by leveraging visual context-powered news. Fake news articles can contain misrepresented, irrelevant, and forged images to mislead the readers (Han et al., 2020; Borchert et al., 2023). The recent

increase in fake news is affecting the social fabric, dynamics, public affairs, policies, and events negatively. Spreading such news through messages is easily achieved through social media by targeting the vulnerable audience who would forward and share the content. Sean Parker, who is Facebook's first president, is strictly critical of social networks and accuses these platforms of being vulnerable to humans (Moravec et al., 2020).

While most of the existing fake news detectors used only one modality (image or text) to identify the suspicious behavior of fake news propagators but one modality is not enough to handle such a complicated problem. Existing content-based fake news detection methods either solely consider textual information (Zhou et al., 2019), or visual information, or combine textual with visual data to analyze the fakeness of a forwarded message without investigating the behavior of fake news propagators (Wang et al., 2018; Yang et al., 2018). Most of the fake news propagators prefer to write fake news with tempting images whose content does not have any relationship with actual news to engage the users and attract the publish attention. Furthermore, when fake news propagators create a fake message with fictional scenarios, it is very difficult for users or any news agency to find the original and tempered images to verify the realness of these messages. Therefore, a research gap exists in the literature between multimodal (text and images) dimensions of misinformation when fake news propagators use real images (non-manipulated) to spread non-factual news on social media channels (Yang et al., 2018). For example, there was a rumor on the internet in 2013, which claimed that United States President Obama was injured in an explosion in the white house, which crashed the share market and abolished over \$150 billion in stock values (Rapoza, 2017). Thus, it has become the greatest threat to economies and freedom of expression. In this scenario, it becomes hard to differentiate between relevant or non-manipulated multimedia and such fictitious news. With non-manipulated images or videos alongside some fictional scenario-based news, there comes a justifying gap between both.

Most of the existing research is still focused on text processing, unimodal features, and using traditional ML techniques to detect misinformation and fake news. With the recent development of advanced ML and deep learning models, there has been less previous evidence for developing multimodal fake news detectors. This is a significant and important gap; given that misinformation, disinformation, and fake news are becoming multimodal in nature in the past couple of years (Begley, 2017).

In recent times, advanced ML and deep learning models have been in trend and are known to achieve excellent ML accuracy in various fields like computer vision, speech recognition, natural language processing, social network filtering, audio recognition, etc. It has shown tremendous results in tasks such as image captioning, visual question answering, and fake news detection. A model was proposed by Jin et al. (2017a) that extracts the visual, textual, and social content features, which are further fused by using the attention mechanism. The Attention mechanism is an effort mechanism by deep neural networks to concentrate on selective or relevant features- while neglecting the others in the network. Another model to learn event-invariant features was proposed, which made use of an adversarial fake news network with a few visual and textual features. However, there remains unclear room for discovering correlations between different modalities and the role of feature engineering to improve the accuracy of the fake news detection models, which is not covered by these models as well. This leads us to propose a novel multi-modal framework for fake news detection. The proposed model takes into account the two modalities present in an article – text, and images. Forging images is one of the most popular ways to tamper with the credibility of news and recent studies have shown that visual features (images) play a very important role in detecting fake news (Wu et al., 2015; Zhang et al., 2019). Rubin et al. (2015) use several ML techniques to analyze the textual data to classify a news item as real or fake but they only used one modality-text to develop the detecting model. Though important, but multimedia content has not been

explored much in the context of fake news detection. Although, basic features of images have been explored by [Tian et al. \(2013\)](#) but these features merely depict the complex distributions of image content and are handcrafted. Therefore, in order to improve the fake news detection systems, the complex relationship between different modes, the role of feature engineering, and varying distributions need to be addressed.

Feature engineering and explainable AI are two interconnected concepts and this connection lies in the interpretability of features. When feature selection is done effectively, it can lead to more interpretable and explainable ML models. By selecting and constructing important features that are capable to capture relevant information from the data, it becomes easier to understand the decision-making process of the ML model, and this process could enable better transparency and interpretability. In summary, feature engineering plays an important role in the development of an explainable AI system by shaping the features used in the ML model, which in turn affects the model's interpretability ([Ali et al., 2023](#)). In this paper, in order to effectively identify the disinformation we develop a multimodal fake news detector model, which extracts textual and visual features particularly to detect the disinformation on benchmark multimodal datasets. We extract several novel features e.g., content-based (topics of an image, contrast, brightness, the dominant and fraction representation of color spots in the manipulated area, etc.), emotions-based (sentiment polarity, adult content, violent content or smile, etc.) and manipulation-based (spoofed content, image sharpness/blur, pixel density difference in the manipulated area, etc.). There have been several efforts at developing fake news detection models in recent years ([Horne & Adali, 2017](#)). Overall, we found that no published literature that considers specific aspects of feature engineering in developing the multimodal fake news detector has been published to identify misinformation and fake news propagators. To address these research gaps, we aimed to develop a fake news detector model to detect the behavior of fake news spreaders using an innovative modified JAYA feature optimization model.

In addition, to developing the detector model, we also addressed the following research questions to investigate the behavior of fake news propagators:

RQ-1: From the psychology perspective, do fake news propagators use a specific linguistic tone or write the news under a certain rule of the press (e.g., fewer words than real news, more question marks, exclamation, use third-person pronouns more than first-person pronouns and capital letters than real news, etc.)?

RQ-2: From the cognitive perspective in the text features, do fake news writers use exclusive words and negations more frequently than real news writers?

RQ-3: From the emotional perspective in the text features, do fake news propagators use more anger, sexual, and swear words and use more lexical diversity to write the news?

RQ-4: From the emotional perspective in the visual features, do fake news propagators use more artificial text and violence to promote violent extremism?

RQ-5: How do multimodal fake news detection methods perform compared to unimodal methods after combining the visual features with text features?

RQ-6: How do feature engineering methods improve the classification accuracy of the proposed multimodal fake news detector model?

This research work conducts several experiments on three datasets collected from popular social media platforms: Twitter, Weibo, and rFakeEdit, which use textual and visual analytics to extract the important features. The main innovation of our research work is to use the power of textual and visual analytics with an optimization feature engineering algorithm that selects the optimal features to develop the multimodal fake news detector model.

2. Literature review, conceptual background, and theory building

In this literature review and background sections, we discuss the published works related to the use of feature engineering for developing fake news detector models from single-modality and multimodal categories. In the first part, we discuss several theories to extract useful features from the multimodal data and to make theoretical ground for developing a fake news detector model using these features.

Johnson and Kaye, stated that most people specifically use social media for hedonic purposes, connecting with friends and seeking entertainment, rather than for anything that might be of use. When using social media, the user does it with a different mindset than when reading news items found elsewhere on the internet. This difference in the consumption of information affects how the user processes information. As an example, it is known that some product reviews are fake but users do not read them for entertainment, they only read them in order to make an informed choice as to whether to buy an item, knowing there is a monetary incentive when making the best decision. For this reason, users reading fake reviews have a purpose in mind, having a utilitarian mindset, and the goal is to understand the content of the review and whether the information should be considered in making a decision. A paper by Minas et al. investigated the utilitarian mindset in a decision-making context within virtual team interactions. Participants involved in a decision-making team-based chat were found to use confirmation bias. Contrastingly, when reading social media news, the user has a hedonic mindset and the goal is pleasure and enjoyment, not one where they have to decide whether the content is fake or not. The user wants to avoid any activity that feels like work, e.g., thoughtful information processing, and to avoid anything that they do not enjoy, e.g., reading stories where their favorite team has lost. [Moravec, Minas and Dennis \(2019\)](#) state that users want to read feel-good articles, i.e., ones that make them feel happy and that often support what they believe. [Moravec, Minas and Dennis \(2019\)](#) go further to say that on social media, the source of the news is not always clear, whereas when a user is looking at traditional sources for news on the internet, they will have to visit their favorite news network or newspaper online which they would consider trustworthy and would therefore have an understanding of the source's limitations. Facebook differs in that it uses an algorithm to choose the articles it publishes and they are not the user's choice ([Yfanti et al., 2023](#)).

Users may subscribe to some information sources by following them on social media platforms, but many sources of information come in the forms of advertisements, shares by friends, and decisions made by the algorithms employed. Therefore, there is a mixture of different sources, reputable and disreputable. For example, a fake news item may appear between a CNN article and an advertisement for Aunt Martha's cookies! [Kim and Dennis \(2018\)](#), and [Moravec et al. \(2019\)](#) both concluded that if the source of the item is not clear, or is even deliberately hidden, users with a hedonic mindset will not bother to make the effort to find and understand the source.

Fake news appears everywhere and in vast amounts resulting in it being hard to separate fact from fiction. [Silverman \(2016\)](#) pointed out that more fake news than real news is shared on social media. [Moravec, Minas and Dennis \(2019\)](#), state that the low cost and massive presence of fake news is a reason why it is so common on social media. They also claim that many fake sites have appeared on Twitter and Facebook solely to destroy a specific individual or to spread propaganda, carefully crafted for a specific reason. The hedonic mindset, not knowing the source and, the sheer volume of fake news, are three reasons, combined to form three contextual factors, why users do not think critically as they otherwise would when presented with news from a known reputable source. [Gabiello et al. \(2016\)](#) proved further that more than half of the items shared on Twitter have not been completely read in the first place, and the users have not thought about the content in a critical manner.

In the 1980s, two complementary models were proposed for

measuring the cognitive process. These were Elaboration Likelihood Model (ELM), Heuristic-Systematic Model (HSM), proposed by Chaiken (1980), and Chaiken and Eagly (1983), respectively. The basic fundamental concept of HSM and ELM was common and both models discuss two different cognitive processes, which form cognitive attitudes, but differ in cognitive processing, which is used in information evaluation. These are a quantitative difference and a qualitative difference, respectively. HSM and ELM argue that the route that is chosen by an individual is based on the motivation and ability to participate in extensive cognition. Both models have evolved to say that cognition is a continuum of processing and that the routes are not separate. The most popular of the two is ELM and it is widely used today (Cacioppo et al., 2018; Moravec, Minas & Dennis, 2019). That is why we decided to use ELM and framing theory concepts to identify the text and visual features in this research.

Petty & Cacioppo, used the ELM model to show that online users can be influenced whether to accept social media channels' information as true either by using the peripheral (heuristic) or central processing route. The individual needs to take time and cognitive effort to rationally and objectively assess the truth of the item. It relies on people having emotions triggered and requires little cognitive effort. Rational judgment and assessment are often neglected when a news item, for example, uses specific language or tone with images that could trigger certain emotions or feelings: anxiety, violence, and fear. Machine-learning models need to take into account the emotions for developing the fake news detector models and the ELM model is useful in identifying how the peripheral processing route is targeted by text and images for fake news items. However, in order to get the true credibility of an article it is important to integrate images, videos, and text together. In 2005, De Vreese presented the *Framing Theory* which states that an issue or event is defined by the way it is presented, (De Vreese, 2005). In framing theory, a frame consists of a set of ideas which are key, stock phrases, and images that support a specific event interpretation. The interpretation made with text and images is a dominant interpretation created strategically and it is easier to understand, more memorable, and easy to accept. The combination of images and text makes it easy to effectively manipulate ideas and give deniability to the website or the author. It is also hard for automated systems to identify fake news or misinformation if they rely only on text and images (Messaris & Abraham, 2001). In the work presented here, many theories are used in developing the fake news detector model. Lang used the LC4MP model that enables the understanding of how information-rich, multi-modal messages are processed. Apparently, there are several mechanisms that could determine that a person will register cognitively his profile on social media websites and share fake news consciously or unconsciously. It is, therefore, important that the deep learning and advanced machine-learning models that are used in the current work identifies fake reviewers and captures several media channels of information so that the salient features within a modality are processed as far as possible.

In data science, feature engineering or feature selection (*selecting the best features for building the ML models*) is the most important data pre-processing challenge that is used to reduce the dimensions of the datasets by removing unimportant, noisy, and irrelevant features or variables (Kumar et al., 2022). Generally, we classify the feature selection methods into two categories: wrapper, and filter-based techniques. Filter-based methods (LDA, ANOVA, Chi-square, etc.) are used to rank the variables based on the scores of several statistical tests. All Filter-based methods use the similarity methods (fisher or Laplacian score), and statistical methods (F-score, Gini index, or Chi-square statistic score) to select the best features for building ML models based on specific criteria. One of the major issues with the Filter-based methods is that method does not remove multicollinearity (Kumar et al., 2022). In wrapper-based methods, we use the inferences that we have drawn from

the previous ML model and evaluate the quality of variables based on this score. For adding or removing the variables in the model, this method first searches the best subset of variables and after that, the model evaluates the fitness of the subset of variables before developing the ML model building block (Kumar et al., 2022). The Wrapper-based methods are still unexplored and researchers are developing new methods using feature optimization techniques these days. Several wrapper-based feature selection methods have been developed by the researchers with their own challenges. Too et al. developed the feature optimization algorithms to find the best subset of suitable features for developing the ML models. All these feature optimization algorithms are developed by specific controlling parameters such as weights, number of iterations, crossover probability, etc. that need to be tuned to get the best subset of optimized features. To achieve good accuracy in the ML building process, parameter tuning is essential and a good choice of parameters can make the ML model succeed after getting the best subset of important features (Peng & Xintong, 2022).

Recently, deep learning models have been used to develop fake news multimodal detector models. Most of the published literature (Singhal et al., 2019; Shah & Kobti, 2020; Peng & Xintong, 2022; Uppada & Patel, 2022) used convolutional neural network (CNN), and Visual Geometry Group (VGG)-19 deep learning architectures that use the attention-based mechanism and combines it with other ML model - long short-term memory (LSTM) to develop a microblogging detector model for identifying the characteristics of fake news propagators. The published literature used the CNN model to extract several emotions and manipulation-based fine-grained features and used the LSTM for coarse-grained feature extraction and after that, these extracted features are classified into fake or credible news stories to achieve better accuracy of the misinformation detector models.

However, all of the above detection methods (unimodal and multimodal) have delivered promising results but we observed several limitations in the published literature for identifying the behavioral characteristics of fake news propagators using feature engineering. The multimodal nature of a dataset is always a challenge for researchers for extracting useful features for building ML models. Feature engineering (selecting the best features/variables) or data pre-processing is the most important and critical step in developing fake news ML detector models. Although a few researchers have used feature engineering on the unimodal dataset and developed JAYA and modified JAYA algorithms (Rao, 2016; Das et al., 2020) for finding an optimal feature set of best variables to detect fake news propagators, feature engineering remains uncommon in multimodal dataset modeling. In addition, to the best of our knowledge, a few studies have used unimodal text datasets with preexisting feature sets to detect the behavior of fake news propagators, accounting for feature engineering in multimodal data remains insufficiently explored. Our proposed fake news detector model also differs from other published literature by focusing on investigating the behavior of fake news propagators on psychological, emotional, and cognitive perspectives in a hierarchical manner using the optimized set of best features. Singh et al. (2017) research could be considered the first step towards understanding the behavior of fake news propagators on multimodal datasets; however, a number of questions regarding feature engineering to improve the accuracy of fake news detector models remain to be addressed. Similarly, Kumari and Ekbal (2021), Zhang et al. (2022) contributed to the multimodal fake news detection literature mainly focused on developing the model using deep learning but they devoted less attention to identifying the behavior of fake news propagators and improving the accuracy of the detector with feature engineering tasks. Thus, the use of feature engineering in predicting the behavior of fake news propagators and developing the multimodal fake news detector model is a promising area of research. The main contribution of our research study is summarized below.

- 1 To the best knowledge, we develop an approach that investigates the behavior of fake news propagators and establishes the relationship between textual, and visual features in predicting misinformation & fake news;
- 2 We propose a novel multimodal (with text and image features) fake news detector model to examine the behavior of fake news propagators through psychological, cognitive, and emotional perspectives; and
- 3 After developing the fake news multimodal detector model, we conduct several experiments on real-world datasets to prove the effectiveness of our proposed model.

3. Feature design, data description, and fake news detector model development

The theoretical concepts, Framing Theory, LC4MP, and ELM, described in the last section, helped us to identify the features for developing the fake news detector model. Five broad sub-categories (user-based, content-based, style-based, emotions-based, and manipulation-based) were identified into two categories (text and visual) from this review:

3.1. Text-based features

Linguistic approaches that are used to detect news, which is fake generally, rely on the use of language and its analysis (Singh, Ghosh & Sonagara, 2020). There are two user-based features: user profiling features and user credibility features. User profiling features (account name, geolocation information, registration information, verified, not verified (Zhang & Ghorbani, 2019), has a description and does not have a description) and user credibility features (user's credibility score (Chu et al., 2012), number of users' friends and followers, the ratio between friends and followers, number of tweets post/tweets (Castillo et al., 2011; Zhang & Ghorbani, 2019) have been used in the investigation of suspicious user and non-human accounts by capturing their unique characteristics. Content-based features correspond to the Framing Theory's Issue Selection and central persuasion route for ELM. LIWC-based categories of textual content were used in accordance with the work done by Horne and Adali (2017). Tausczik and Pennebaker give an extensive list of LIWC-based features. LIWC contains five main categories and a few sub-categories, e.g., social, cognitive, affective, and perceptual (Pennebaker, Mehl & Niederhoffer, 2003).

Style-based features correspond to the peripheral persuasion route for ELM. In order to analyze the overall complexity of fake messages or stories, punctuation (e.g., question marks, exclamation points), quotes, negations (e.g., no, never, not), and grammar are used (Singh et al., 2020). The framing theory and previous research is done to investigate fake news to accentuate the importance of the relatability and impression of the overall article, Horne and Adali (2017). Ahmed found that fake writers take longer to write articles and make more mistakes because of the timespan features, the namely the average interval between words, average time span of a word, timespan of document, and editing pattern features, such as a number of deletions, arrow keystrokes and "Mouse-Up"s. In the Emotions-based features, Messaris and Abraham (2001), point out that a major determinant of the quality of information is considered by the ELM to be sentiment polarity, Osatuyi and Hughes (2018). LIWC dictionaries are used to measure the effect of words, sentiment score, and SentiStrength which are important measures to identify the intensity of positive and negative emotions (Dickerson et al., 2014; Horne & Adali, 2017; Zhang & Ghorbani, 2019).

3.2. Visual features

Images require a smaller cognitive load than text and are not as

intrusive; therefore considered powerful framing tools. This means that the peripheral route in the brain is activated and the users are much more interested to see the visual frames in their mind without questioning it (Singh et al., 2020). Identification of a fake news item using content-based features can be made by examining the image content for clues. It corresponds to Framing Theory's Issue Selection and the central persuasion route for ELM. Features identified can be objects in an image, and the presence and number of faces in an image. Other indicators of fake news are an analysis of color components and properties of images. Style-based features discuss that Fake news can also be identified from features such as size, width, etc., of the image (Blei, Ng & Jordan, 2003; Papadopoulou et al., 2017), and it corresponds to the issue salience component for framing theory and peripheral persuasion route for ELM. In the Emotions-based features, emotions and expressions can also be portrayed by the faces in the image and the presence of violence and so these are also included as features. Pantti and Sirén (2015) showed that when fake news is compared to credible news, it is more visually striking and eye-catching. The images also tend to show accidents, abuse, injury, conflict, and other disturbing material (Jin et al., 2017b).

Manipulation-based features in visual features correspond to the peripheral persuasion route for ELM. Lin et al. (2009) showed that any tampering or manipulation of the image in an article is very important for the determination of fake news. Jin et al. (2017b), Zhang and Ghorbani (2019) investigated visual-based features, namely: similarity distribution histogram, image ratio, clarity score, diversity score, long & hot image ratios, clarity score, and coherence score for identifying the fake news propagators from the data sets. Castillo et al. (2011) shows that the statistical features of an image can also be useful in detecting fake news. The *imagetweet ratio*, from basic image statistics, is already used as a feature for distinguishing fake news, Jin et al. (2017b). One of the most important inspections of images is that most images have a standard resolution, while others do not. Therefore, the different types of tweets can be described in two ways- popularity (hot image) and long image. A hot image is the news event's most popular tweet and popularity is the number of re-tweets of the hot image. A long image is an image with a length-to-width ratio that is greater than 1.9. Generally, it is composed of several different images. The summary of textual and visual features used in the research can be seen in Table 1.

3.3. Data description and model development

One of the most important challenges of building fake news multimodal detector models is obtaining news items that are clearly classified as fake or real. We used several real-world multimodal datasets (Twitter, Weibo, and rFakeEdit) to test our proposed detector model, which has several textual and visual behavioral characteristics of fake news propagators.

We used the Twitter dataset for developing, and testing the fake news detector model that was released by Boididou et al. (2015) as part of a data science challenge (MediaEval, 2015), and the aim of the challenge was to detect the misinformation & fake news content on Twitter website. The Twitter dataset is publically available and has text (in the form of tweets), visual (in the form of images), and additional social contextual dimensions of users. The Twitter dataset encompasses around 18,000 multimodal tweets, of which 10,000 are fake news and 8000 belong to the real news category. In the data pre-processing stage, we deleted a few tweets or messages from the original dataset that do not have any text or image because we are focusing on developing a multimodal detector tool. The other dataset, we used to test the accuracy of our proposed fake news detector model was the Weibo dataset, which is also publically available and used in this research work (Wang et al., 2018). In the Weibo dataset, authors collected the multimodal dataset from the official Chinese news agency (Xinhua agency) and released this dataset to track the misinformation and behaviors of fake news

Table 1
Summary of features (textual and visual) used in developing the fake news detector model.

Feature's Classification	Theoretical Research Support	Sample Text Features	Sample Visual (Image) Features	Empirical Literature Support
User-based Features	Central route processing with ELM framing issue	Username length, number of tweets, has personal URL, is account verified, friends count, followers count, follower-friend ratio, the time listed (number of times the user has been listed/tagged)	Number of hot, and long images in a tweet, popularity score of users (number of shares, re-tweets, and comments obtained by the multimodal tweet)	Singh et al. (2020) , Castillo et al. (2011) , Jin et al. (2017b)
Content-based Features	Framing: Issue selection ELM: central route processing	Sentistrength, topics for personal concern like-work, home, money, religion, number of external links, tags, and URLs in a tweet, social words like- friends, male referents, family, female referents	Topics of image labels, fraction representation of colours, contrast, dominant color in the focused area, celebrity presence, brightness, gender, number of faces in image	Hong (2013) , Shu et al. (2017b) , Pérez-Rosas et al., (2017) , Singh et al. (2020)
Style-based Features	Peripheral route processing with ELM framing issue	Word or sentence level of features, Average length of tweet, average sentence length, sentence complexity, use of punctuation, use of these types of negations (e.g., no, nope, not, never, fake, etc.), slang terms (e.g., lol, brb, etc.)	Multi-image and hot image ratio, number of images, long image ratio	Castillo et al. (2011) , Hong (2013) , Conroy et al. (2015) , Rubin et al. (2016) , Jin et al. (2017b) , Horne and Adali (2017) , Singh et al. (2020)
Emotions-based Features	ELM: peripheral route processing using LC4MP	Average polarity of sentence, swear words, affect words, anger words, assent, emotional tone, sadness, anxiety related words	Adult content in image, violent content in image, blood content or any other medical content, emotions portrayed in image	Gupta et al. (2013) , Jin et al. (2017b) , Singh et al. (2020)
Manipulation-based Features	Issue salience Framing	Discrepancy, certainty, number of impersonal and personal pronouns, tentativeness,	Spoofed content, blurred ratio, multi-image and hot image ratio, image ratio II, image sharpness score, long image ratio	Farid (2006) , Lin et al. (2009) , Pantti and Sirén (2015) , Jin et al. (2017b) , Horne and Adali (2017) , Singh et al. (2020)

propagators in the China region. The authors scrapped this Weibo dataset from 2012 to 2016 and verified it with the Weibo official rumor debunking system.

[Fig. 1](#) presents the four different phases of fake news detector model development: (i) visual data processing using Google vision deep learning architecture, and textual data processing using LC4MP; (ii) textual and visual feature extraction and adding new variables in the original dataset; (iii) modified JAYA feature engineering algorithm development for identifying the important features; and (iv) fake news detector model development. This detector model receives multimodal textual and visual data and the relevant information of Twitter users or news propagators as input.

For the first task, we used the LC4MP framework to extract several textual features (see [Table 1](#)) from the original dataset and after that, we used the Google vision deep learning architecture to extract a wide variety of visual features from the images that are associated with the text data. In the next stage, we perform feature engineering and develop a novel M-JAYA algorithm to select the important features for ML model building and remove the noisy and irrelevant features from the dataset. After that, we develop several ML models for identifying the fake news propagators and use the majority voting concept to select the best three ML models based on their rank of classifiers' performances. We use a simple rule in the majority voting section, and output label Y could be predicted based on three top ML hyperplanes (output = mode{M1(Z), M2(Z), M3(Z)}). We can understand it with this example, if the proposed fake news detector model selects the best three ML models based on accuracy and if model_1 classifies the instance as fake, model_2 classifies the same instance in the fake category, and model_3 classifies it in the real news propagator category, then the detector model uses the majority voting concept and provide the final output as a fake class $Y = \text{mode}\{1, 1, 0\} = 1$. This is the final output of our proposed detector model.

4. Feature engineering, M-JAYA algorithm development, experimentation, results, and discussion

Based on the data set, we had 11,498 news with 6643 of them identified as fake news and 4864 of them identified as real news. We

accessed them in different features including language features, word features, linguistic features, grammar features, affective processes features, text search, manipulation techniques, objectives, and visual features. The general null hypothesis was set that for each feature the fake news was not statistically different from real news, i.e. fake news propagators did not use any specific features or techniques to produce the fake news. To check the details of the difference between fake news and real news based on the effect of the feature, the list of null hypotheses from research questions with corresponding features were developed in [Table 2](#).

To give the answer to the RQ1, RQ2, RQ3, and RQ4, and to test the significance of the hypotheses, we used several t-tests to test the difference in the language features between fake news and real news in the data set. [Tables 2–7](#) presented the mean and standard deviation for each variable as well as the p-values.

From [Table 2](#), we found that the P value was less than 0.05 for the language features (analytical thinking words, clout words, authentic words, and emotional tone). Therefore, the null hypothesis was rejected, and the conclusion was presented that the fake news propagators write the news using a specific linguistic tone, i.e., using less analytical thinking words, fewer clout words, less authentic words, and less emotional tone. From [Table 3](#), we found that fake news propagators write the same count of total words and dictionary words as real news writers. Also, the fake news propagators write fewer sentences, but more big words (words larger than 6 letters) than the real news. From [Table 4](#) results, we found that fake news propagators write the news using negations (e.g., 'no', 'not') more frequently than real news writers, but use conjunctions less than real news writers. From [Table 5](#) results, we concluded that both fake news and real news use more first-person pronouns but there was no difference in the use between fake news and real news. From [Table 6](#), the conclusion was presented that fake news propagators use more verbs, but fewer comparisons, interrogatives, numbers, and quantifiers than the real news. From [Table 7](#) results, we found that the P value was less than 0.05 for the use of anger, sexual, sadness, and swear words. Therefore, the null hypothesis was rejected, and the conclusion was presented that the fake news propagators use more anger words and swear words, but fewer sexual words and sadness words than the real news propagators. From [Tables 8](#) and [9](#)

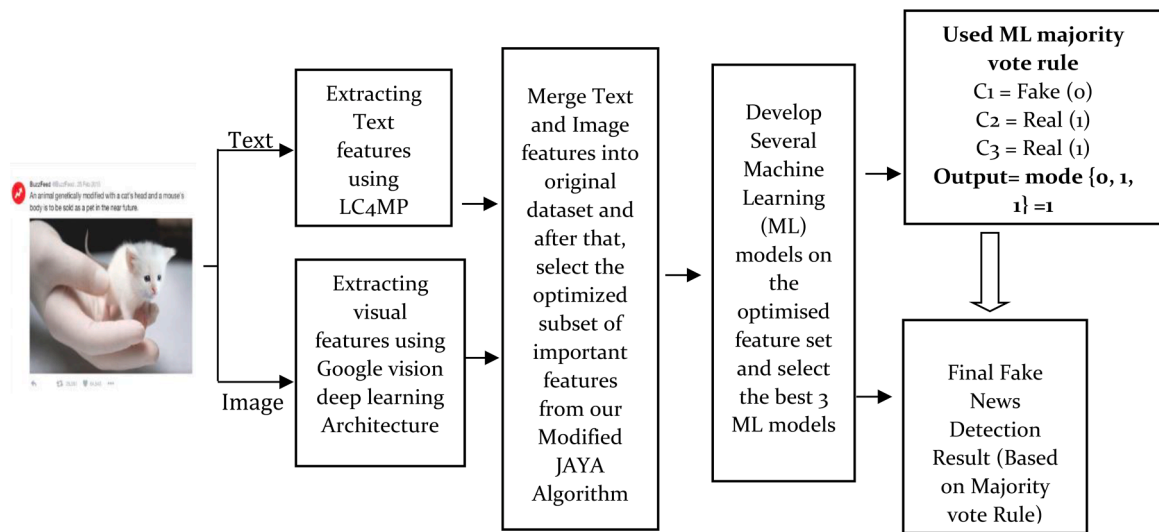


Fig. 1. Proposed Multimodal fake News detector Model.

Table 2

Hypothesis	Features	Test
Fake news propagators use the same linguistic tone as real news writers do	Language Features	<i>t</i> Test
Fake news is produced under the same rule of the press as the real news	Word Features	<i>t</i> Test
Frequency of exclusive words and negations used in fake news is the same as in real news	Linguistic Features	<i>t</i> Test
No difference for use of third- and first-person pronouns between fake news and real news	Linguistic Features	<i>t</i> Test
Fake news propagators use the same lexical diversity as real news writers do	Grammar Features	<i>t</i> Test
The use of anger, sexual, sadness, and swear words in fake news are the same as in real news	Processes Features	<i>t</i> Test
No difference in the use of artificial text and violence between fake news and real news	Text Search	Chi-Square Test
Fake news propagators do not use specific manipulation techniques	Manipulation Technique	<i>t</i> Test
No specific pattern of difference from objective between fake news and real news	Objectives	Chi-Square Test
Fake news propagators use the same visual features as real news writers do	Visual Features	Chi-Square Test

Table 3
Summary Statistics of Language Features by Fake or Real News.

	Total	Fake News	Real News	<i>P</i> Value
Language Features, MEAN (STD)				
Analytical thinking	86.35 (19.19)	85.7 (20.24)	87.24 (17.62)	<0.0001
Clout	54.3 (21.34)	52.21 (20.91)	57.15 (21.59)	<0.0001
Authentic	15.38 (23.49)	14.71 (23.03)	16.29 (24.08)	0.0004
Emotional tone	42.1 (35.57)	40.2 (34.76)	44.7 (36.48)	<0.0001

results, the conclusion was presented that the fake news propagators use more artificial text and violence than real news to promote violent extremism and we can identify patterns in the behavior of fake news propagators by checking the “Object” column, i.e., Animal, Building, Person, Sculpture, and Plants and “object” variable could be an important factor for understanding the behavior of fake news propagators.

In the next phase, we used the features in the univariate and multivariate logistic regression (LR) model to understand the behavior of fake

Table 4

Summary Statistics of Word Count and Language Features by Fake or Real News.

	Total	Fake News	Real News	<i>P</i> Value
Word Features, MEAN (STD)				
Word Count	16.65 (20.65)	16.4 (20.44)	17 (20.93)	0.1247
Words / Sentence	10.68 (5.58)	10.58 (5.69)	10.81 (5.43)	0.0304
Words > 6 letters	22.9 (9.79)	23.37 (10.06)	22.27 (9.37)	<0.0001
Dictionary words	49.36 (14.5)	49.5 (15.05)	49.19 (13.71)	0.2553

Table 5

Summary Statistics of Linguistic Features by Fake or Real News (from cognitive perspectives).

	Total	Fake News	Real News	<i>P</i> Value
Linguistic Features, MEAN (STD)				
Conjunctions	1.02 (2.50)	0.96 (2.51)	1.10 (2.47)	0.0030
Negations	0.62 (2.02)	0.68 (2.14)	0.54 (1.83)	0.0004

Table 6

Summary Statistics of Linguistic Features by Fake or Real News (from psychology perspectives).

	Total	Fake News	Real News	<i>P</i> Value
Linguistic Features, MEAN (STD)				
Difference of singular use	−0.45 (2.14)	−0.48 (2.31)	−0.40 (1.89)	0.0502
Difference of plural use	−0.04 (1.56)	−0.02 (1.61)	−0.07 (1.50)	0.0601

news propagators and to evaluate the effect of each feature to develop the fake news detector model. We used the GEE model (Generalized Estimating Equations) to process the analysis. The importance of the GEE model is that it produces efficient estimates of all coefficients by taking the over-time correlations into account when producing the estimates and it will typically be a block-diagonal matrix. The models were fitted to check the effect on the outcome, and the fake news detection rate, from each dependent variable in different categorical features. We also identified whether the effect varies by each variable throughout the feature. For this, we used multivariate logistic regression to control for the specific effect of each variable in the feature, which is allowed by the panel dimension of our dataset, and estimate the following equation for the detection rate of fake news.

$$P(\text{FakeNewsDetection}) = \alpha + \beta_1 \text{Language} + \beta_2 \text{Word} + \beta_3 \text{Linguistic} + \beta_4 \text{Grammar} + \beta_5 \text{Process} + \beta_6 \text{Text} + \beta_7 \text{Manipulation} + \beta_8 \text{Objective} + \beta_9 \text{Visual} + \mu_i + \epsilon_i$$

The outcome variables were identified as the probability of news being detected as fake news. The independent variables included language features, word features, linguistic features, grammar features, process features, text search, manipulation techniques (copy-move techniques and splicing techniques), objectives, and visual features. Error term, μ_i included an idiosyncratic error term, ϵ_i and fixed effect (μ_i). The idea is that if given news was with some specific features, i.e., specific linguistic tone, certain rules of the press, more lexical diversity, or particular objectives, etc., we would expect to see the rate pattern of detection as fake news. Therefore, β_1 through β_9 capture the average effect of the different features on this pattern. Table 10 presented the output of the model below.

Table 7
Summary Statistics of Grammar Features by Fake or Real News.

	Total	Fake News	Real News	P Value
Grammar Features, MEAN (STD)				
Common verbs	5.45 (6.06)	5.67 (6.33)	5.15 (5.66)	<0.0001
Common adjectives	8.89 (6.08)	8.89 (6.44)	8.90 (5.54)	0.8930
Comparisons	0.72 (2.15)	0.63 (2.02)	0.85 (2.31)	<0.0001
Interrogatives	0.46 (1.65)	0.43 (1.67)	0.50 (1.62)	0.0262
Numbers	0.90 (2.95)	0.81 (2.66)	1.04 (3.30)	<0.0001
Quantifiers	0.43 (1.59)	0.36 (1.48)	0.52 (1.73)	<0.0001

Table 8
Summary Statistics of Affective Processes Features by Fake/Real News (emotional perspectives).

	Total	Fake News	Real News	P Value
Affective Processes Features, MEAN (STD)				
Anger words	0.57 (2.18)	0.61 (2.28)	0.50 (2.04)	0.0113
Sexual words	0.14 (1.04)	0.12 (0.94)	0.17 (1.17)	0.0026
Sadness words	0.22 (1.20)	0.15 (0.95)	0.32 (1.46)	<0.0001
Swear words	0.64 (2.41)	0.71 (2.53)	0.54 (2.23)	0.0003

Table 9
Summary Statistics of Text Search by Fake/Real News (emotional perspective in visual features).

	Total	Fake News	Real News	<i>P Value</i>
Text Search: Adult, N (%)				
Very unlikely	11,188	6412 (96.7 %)	4776 (98.2 %)	<0.0001
Unlikely	307	222 (3.3 %)	85 (1.7 %)	
Possible	3	–	3 (0.1 %)	
Text Search: Spoof, N (%)				
Very unlikely	9498	5227 (78.8 %)	4271 (87.8 %)	<0.0001
Unlikely	1142	963 (14.5 %)	179 (3.7 %)	
Possible	382	310 (4.7 %)	72 (1.5 %)	
Likely	101	76 (1.2 %)	25 (0.5 %)	<0.0001
Very likely	375	58 (0.9 %)	317 (6.5 %)	
Text Search: Medical, N (%)				
Very unlikely	10,965	6123 (92.3 %)	4842 (99.5 %)	<0.0001
Unlikely	533	511 (7.7 %)	22 (0.5 %)	
Text Search: Violence, N (%)				
Very unlikely	8887	4735 (71.4 %)	4152 (85.4 %)	<0.0001
Unlikely	2607	1898 (28.5 %)	709 (14.5 %)	
Possible	4	1 (0.1 %)	3 (0.1 %)	
Text Search: Racy, N (%)				
Very unlikely	9601	5426 (81.8 %)	4175 (85.8 %)	<0.0001
Unlikely	1677	1066 (16.1 %)	611 (12.6 %)	
Possible	55	27 (0.4 %)	28 (0.6 %)	
Likely	139	89 (1.3 %)	50 (1.0 %)	<0.0001
Very likely	26	26 (0.4 %)	–	

Based on the model output, the predicted probability plots of fake news detection are shown in Fig. 2, we can conclude that

- Fake news detection was significantly affected by language features. The decrease of analytical thinking words, clout words, authentic words, and emotional tone would increase the possibility of the detection of fake news. In addition, the clout words had more effect than other words.
- From the word features, sentences had a positive effect and big words (words larger than 6 letters) had a negative effect on the fake news detection rate. The strengths of the effect were almost the same. But dictionary words did not have any effect on the detection rate.
- For the linguistic features, if negations would be found 20 % more in a news than others, the probability of this news being detected as fake news would be increased by 15 %–20 %. In addition, if news found 20 % more conjunctions, the probability of fake news detection would be decreased by 10 %.
- By increasing the detection rate of 10 % for the grammar features (including verbs, comparisons, interrogatives, numbers, and quantifiers), fake news detected possibility varied from less than 5 % to around 20 %. Among these variables, quantifiers had the largest effect. In other words, more quantifiers in a news would make less possibility of fake news.
- Anger words and swear words had a positive effect, and sexual words and sadness words had a negative effect on fake news detection. All the effects were significant.
- Fake news detection rate was significantly affected by image ratio. If this ratio was increased by 60 %, the probability of fake news detection would be increased by 20 %, which means that the copy-move technique played an important role in the process of fake news producing.
- For the splicing technique used in producing fake news, the effect from the visual score (visual clarity score, visual clustering score, visual coherence score, and visual diversity score) reflected this situation. A lower score indicated a higher rate of fake news detection.

Table 10

Multivariate Generalized Linear Mixed Model for Fake New Detection by each Feature through Logistic Regression.

	aOR (95 %CI)	P-Value
Total Number of News	11,498	–
Language Features		
Analytical thinking	0.996 (0.994,0.998)	0.0003
Clout	0.989 (0.988,0.991)	<0.0001
Authentic	0.996 (0.994,0.997)	<0.0001
Emotional tone	0.997 (0.996,0.998)	<0.0001
Word Features		
Words / Sentence	0.991 (0.984,0.998)	0.0078
Words > 6 letters	1.014 (1.010,1.018)	<0.0001
Dictionary words	1.005 (1.002,1.007)	0.0006
Linguistic Features		
Conjunctions	0.976 (0.961,0.990)	0.0011
Negations	1.038 (1.018,1.058)	0.0002
Grammar Features		
Common verbs	1.020 (1.013,1.027)	<0.0001
Common adjectives	1.005 (0.999,1.012)	0.1248
Comparisons	0.954 (0.937,0.971)	<0.0001
Interrogatives	0.960 (0.938,0.982)	0.0005
Numbers	0.975 (0.963,0.988)	0.0001
Quantifiers	0.943 (0.921,0.966)	<0.0001
Affective Processes Features		
Anger words	1.002 (0.974,1.031)	0.8927
Sexual words	0.893 (0.856,0.930)	<0.0001
Sadness words	0.888 (0.859,0.917)	<0.0001
Swear words	1.053 (1.025,1.081)	0.0002
Copy-move technique		
Image Ratio	1.012 (1.008,1.016)	<0.0001
Long Image Ratio	1.000 (0.996,1.003)	0.8277
Hot Image Ratio	0.964 (0.953,0.974)	<0.0001
Image Sharpness	1.030 (1.015,1.044)	<0.0001
Blurred Ratio	1.017 (1.013,1.021)	<0.0001
Splicing technique		
Visual Diversity Score	0.993 (0.991,0.995)	<0.0001
Visual Clarity Score	0.988 (0.987,0.989)	<0.0001
Visual Coherence Score	0.958 (0.955,0.961)	<0.0001
Visual Clustering Score	1.034 (1.032,1.036)	<0.0001
Test Search		
Adult	0.387 (0.269,0.557)	<0.0001
Spoof	0.559 (0.523,0.597)	<0.0001
Medical	2.255 (1.420,3.582)	<0.0001
Violence	5.061 (4.347,5.892)	<0.0001
Racy	0.907 (0.820,1.003)	0.0571
Objective		
Animal	4.293 (3.321,5.551)	<0.0001
Building	0.861 (0.755,0.982)	0.0257
Person	1.617 (1.466,1.784)	<0.0001
Sculpture	3.223 (1.820,5.708)	<0.0001
Plant	0.942 (0.680,1.304)	0.7175
Visual Feature		
Face of Joy	2.447 (2.038,2.937)	<0.0001
Face of Exposed	1.210 (1.101,1.329)	<0.0001
Face of Blurred	0.415 (0.379,0.455)	<0.0001
Face of Headwear	0.606 (0.514,0.714)	<0.0001

- By checking text search of “Adult”, “Medicine”, “Racy”, “Violence”, and “Spoof”, except “Spoof”, all the other texts had a positive effect on the detection of fake news, which means that fake news propagators use more artificial text and violence to promote violent extremism.
- On the behavior of information identification, if the objectives came to “Sculpture”, “Plant” and “Animal”, the probability of fake news increased by 35 %–45 %; if the objectives were chosen from “Person” or “building” the rate decreased by 15 %–25 %.
- From the last section of the plot, all the visual feature (face of joy, face of exposed, face of blurred, and face of headwear) had a negative effect on the fake news detection possibility, which presented opposite results to our research questions. That was because the sample power was too small. Based on the summary statistics, news without faces shown took place about 90–95 %. Therefore, only around 5 %–10 % of the sample was used to check the effect of visual

features on fake news detection, which was a lack of power. Fake news multimodal messages use a specific manipulation technique, i. e., resampling/copy-move/ splicing. For instance, fake news was detected using a larger image ratio, but less hot image ratio and less image sharpness. In addition, fake news was found with a significantly lower score of visual diversity, visual clarity, visual coherence, and visual clustering.

After understanding the characteristics and behavior of fake news propagators, we started to work on the first phase of any ML project-data processing and feature engineering. One of the main contributions of this research is to build a novel fake news detector framework based on an optimized set of important features. In order to identify the best features and understand the importance of individual variables of the fake news detector model, we tried to control the parameters of the JAYA algorithm and we modified it to design the optimal subset of important features for improving the ML classifier’s performance. The detailed description of the modified JAYA algorithm is described in detail below. In the first step, we initialize JAYA algorithm parameters and four parameters need to be initialized: Population Size (**P**), Number of Executions (**E**), Maximum Number of Iteration (**MaxIter**), and the minimum number of features (**D**). Once parameters are initialized, it is essential to define the training and testing datasets. After that, the algorithm needs the objective function to start the process. This objective function aims to select the optimal subset of the features that minimize the fitness function of the selected classification models. The error is computed as the difference between the actual output, and the selected model estimation is presented by Eq. (1), where $k = 1, 2, \dots, m$ and m is the number of testing observations $m = M_{ma}^e$ for a dataset.

The fitness function values of the model are computed by dividing the summation of errors by the number of observations presented by Eq. (2).

$$Error(k) = (\hat{y}(k) \neq y(k)) \quad (1)$$

For each population p where, $p = 1, 2, \dots, P$, the objective function (Eq. (2)) is calculated using Eq. (1).

$$Fitness(p) = \frac{\sum_{k=1}^m Error(k)}{m} \quad (2)$$

After that, the binary JAYA population is reconstructed using the S-shape transfer function and a selected subset of features and this S-shape function is used to convert the input to binary values (Kennedy & Eberhart, 1997). After that, the updated solution vector $X(d)$ will be assigned into Eq. (3) to calculate its binary vector.

$$B(X_{new}(p, d)) = \frac{1}{1 + \exp(-5 \times X_{new}(p, d) - 0.25)} \quad (3)$$

The binary value from Eq. (3) further needs to label X_{new} into either ‘1’ or ‘0’. Two methods are proposed here. For *Random r*

$$X_{new} = \begin{cases} 1, & B(X_{new}(p, d)) > rand \\ 0 & \text{Otherwise} \end{cases} \quad (4)$$

Here, rand () is the random number distributed between the range of 0 and 1.

Default r

$$X_{new} = \begin{cases} 1, & B(X_{new}(p, d)) > 0.5 \\ 0 & \text{Otherwise} \end{cases} \quad (5)$$

The next step would be to execute the binary JAYA algorithm and construct the initial binary populations. In this step, each binary population is generated using a random selection of features. Later on, the fitness function cost of each generated population is calculated. In the next step, the best and the worse solutions are selected based on the objective function’s value. If it selects (objective function = error function) option, the minimum value will generate the best solution, and

the maximum value will generate the worse. If it selects (objective function = AUC score) option, the maximum value will generate the best solution, and the minimum value will generate the worse.

In the next improvement step, each and every decision variable of all iterations are modified using the JAYA operator formulated in Eq. (6) (Awadallah, Al-Betar, Hammouri & Alomari, 2020).

$$X_{new}(p, d) = x(p, d) + r1 \times (bestX - |x(p, d)|) - r2(worseX - |x(p, d)|) \quad (6)$$

In the next step of the selection process, when a new solution is generated, it compares to the current solution X_p that is stored in the population's p^{th} position. Suppose the fitness value of a new solution X_{new} is better than the current solution X_p and in this case, the new solution will replace the current solution. This new fitness is compared with the old fitness.

- Objective function = error function: If the new fitness value is less than the old fitness value, its corresponding population is selected for the next iteration and otherwise dropped.
- Objective function = AUC score: If the new fitness value is more than the old fitness value, its corresponding population is selected for the next iteration and otherwise dropped.

The last step is the stop criterion. The Second to fourth steps are repeated until either one of the following conditions is satisfied:

- The maximum number of iterations (i.e., Max_itr) are satisfied.

- Only one population is left.

In the next phase, the effectiveness of the modified JAYA algorithm is tested and compared with several other methods on a few benchmark datasets (PIMA, Musk, Sonar, Madelon, Colon, Leukemia, and Vehicle) that are available freely on the UCI machine learning repository (Table 11–13).

5. Method validation and performance comparison with other studies

For methodological validation, we tested our fake news detector model on the other two datasets – Weibo and, r/Fakeddit. We developed and repeated the data pre-processing and feature optimization steps on two new datasets and our proposed model provided better classification accuracy as compared to other studies. Table 14 shows the accuracy and performance comparison of the proposed fake news detector model with other several published works of literature EANN (Wang et al., 2018), MVAE (Khatter et al., 2019), and cultural algorithm-based multimodal (Shah & Kobti, 2020), and observed that all the key learning performance metrics (Recall and Accuracy) of our model are better than other models. In comparison with other fake news detector models, our modified-JAYA model achieves better Recall (95.5 %), and AUC (96.9 %) scores on the Weibo dataset. We have achieved high accuracy and Recall scores on the third rFakeEdit dataset. Over proposed fake news detector model on the Weibo and rFakeEdit datasets gains 4 % and 8 % improvement in accuracy over the cultural algorithm-based model,

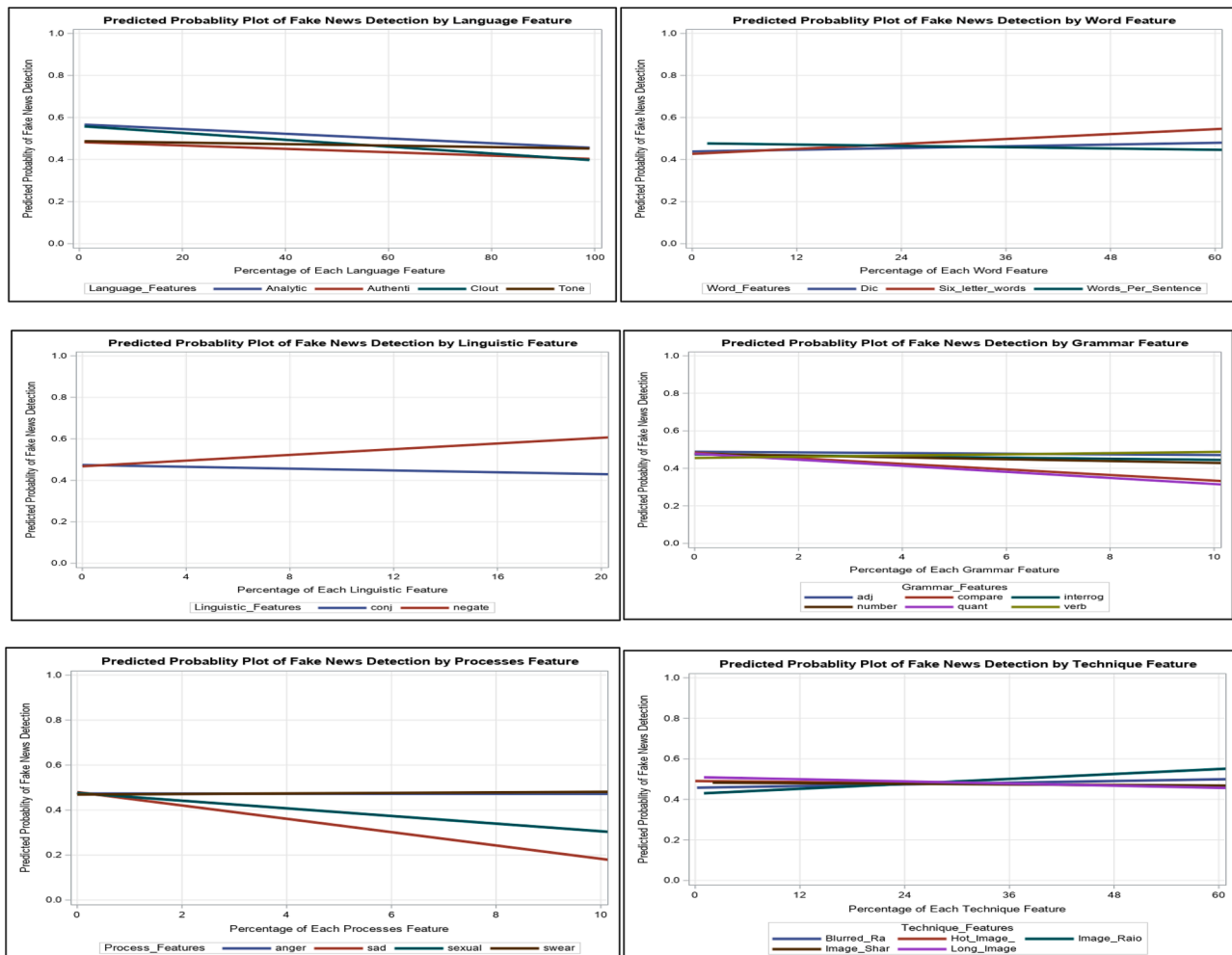


Fig. 2. Predicted Probability Plots of Fake News Detection by each Features.

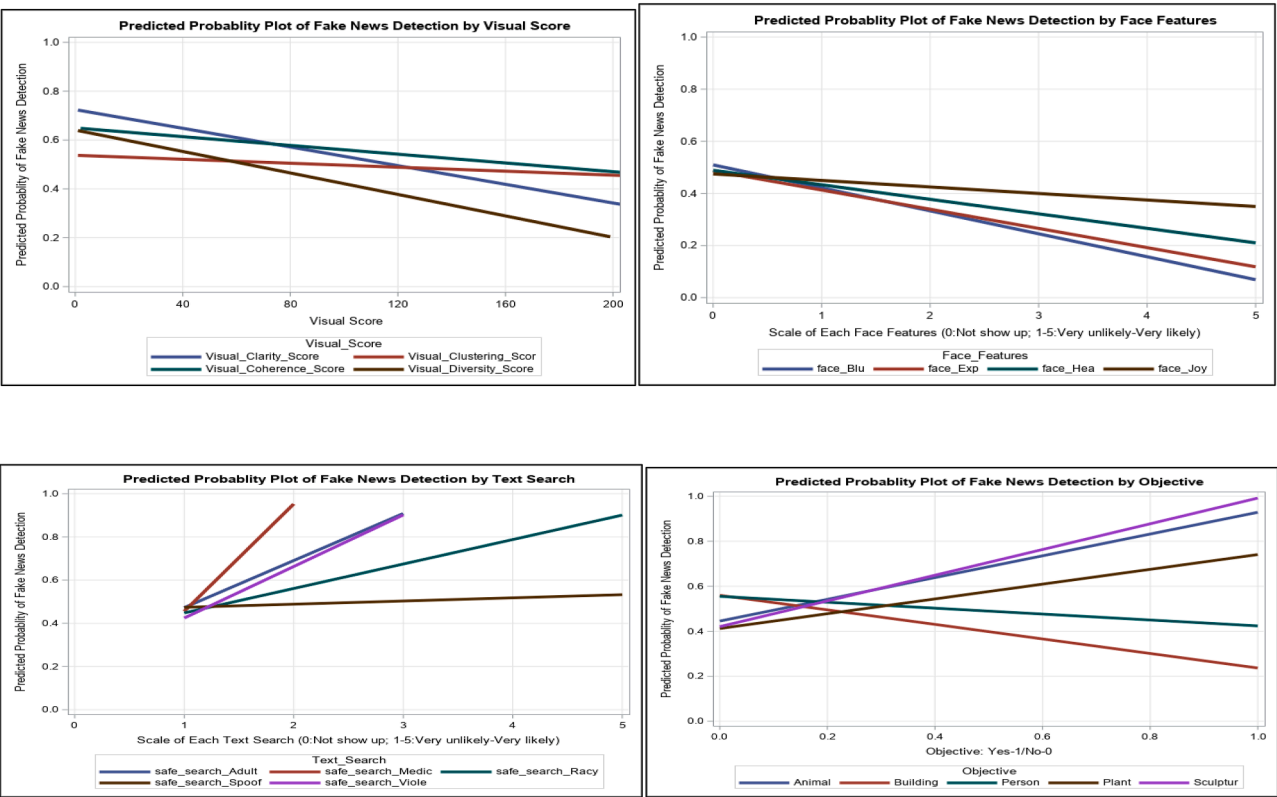


Fig. 2. (continued).

Table 11
Performance comparison of all ML models with and without feature engineering.

Algorithms	Without feature engineering				After using feature engineering by the M-JAYA algorithm			
	Precision	Recall	F-1	AUC	Precision	Recall	F-1	AUC
XGBoost Method	0.716	0.713	0.732	0.713	0.873	0.842	0.882	0.880
LR (Logistic Regression)	0.712	0.696	0.705	0.759	0.804	0.811	0.843	0.856
LSTM	0.693	0.682	0.663	0.747	0.826	0.816	0.811	0.831
ANN	0.670	0.641	0.681	0.731	0.690	0.713	0.742	0.775
RNN	0.641	0.632	0.649	0.736	0.659	0.717	0.729	0.784
RBF- SVM	0.603	0.581	0.592	0.641	0.677	0.688	0.729	0.797
Logistic Regression	0.567	0.734	0.657	0.731	0.739	0.723	0.745	0.781
k-NN (k = 10)	0.556	0.712	0.656	0.723	0.652	0.723	0.744	0.784
Naïve Bayes	0.642	0.591	0.614	0.671	0.689	0.614	0.723	0.741
Random Forest	0.644	0.581	0.589	0.647	0.666	0.681	0.682	0.728

Table 12
Final Model Results.

Algorithms	After using feature engineering by the M-JAYA algorithm (Top 3 ML models)				After applying majority vote rule on feature engineering results $Y = \text{mode}\{0, 1\} = 1$			
	Precision	Recall	F-1	AUC	Precision	Recall	F-1	AUC
XGBoost	0.873	0.842	0.882	0.880	0.932	0.955	0.966	0.969
LR	0.804	0.811	0.843	0.856				
LSTM	0.826	0.816	0.811	0.831				

Table 13
Performance improvement in ML after incorporating visual features.

Algorithms	Precision (%)	Recall (%)	F-1 (%)	AUC (%)
XGBoost method	+8.6***	+3.9***	+6.8 %***	+6.2***
LR	+2.6	+4.6	−2.6	+2.4
LSTM	3.9	+4.8	+6.2	+6.8

*** denote the significance at the 0.001 level.

MVAE, and EANN multimodal models.

6. Conclusion, managerial implications, and future research directions

In this research work, we developed a novel M-JAYA-based fake news detector model that can help social media managers, society, organizations, and online users to detect misinformation and identify the

Table 14
Performance comparison of the proposed fake news detector model with other models on Weibo dataset.

Algorithms	Precision	Recall	F-1	AUC
EANN Model	0.795	0.806	0.795	0.800
MVAE Model	0.824	0.854	0.769	0.809
Cultural algorithm-based multimodal	0.891	0.873	0.822	0.932
Our proposed Model	0.932	0.955	0.966	0.969

suspicious behavior of fake news propagators through social media posts. However several fake news detector models have already been developed, but none of these fully exploits data pre-processing challenges and optimized feature selection, and the characteristics of fake propagators' behavior to understand the fake news phenomenon in new dimensions. Our analysis revealed several new deceptions dimensions in investigating the behavior of fake news propagators. We proved that the decrease of analytical thinking words, clout words, authentic words, and emotional tone can increase the possibility of the detection of fake news. Fake news propagators use more anger and swear words and use the copy-move technique to temper the images in writing the fake news. The fake news detection rate was significantly affected by the image ratio. If this ratio was increased by 60 %, the probability of fake news detection would be increased by 20 %. The effect from the visual score reflected this situation and a lower score indicated a higher rate of fake news detection. We also proved that multimodal analysis can improve the accuracy of the fake news detector model if feature engineering is performed on the dataset before building the model. For the feature engineering task, a new modified JAYA algorithm has been proposed in this research work to find an optimal subset of important features and to improve the accuracy of the fake news detector model. The proposed model selects the best variables/features from the raw dataset and removes the unimportant, noisy, redundant, and irrelevant features that do not contribute to improving the accuracy of the detector model. The experiments and results show that our proposed model is removing the unimportant features effectively and giving better classification results as compared to other traditional feature selection models. The limitation of the proposed feature engineering approach is the high computational cost which is due to searching for the optimal subset of important variables and removing the noisy and irrelevant variables which lead sometimes to the loss of information or important variables.

Developing a fake news & disinformation detection model carries significant managerial implications in businesses, as it involves a combination of technical, ethical, and strategic considerations. A disinformation detection model can improve the decision-making power of managers and it can help to make better decisions by providing them with accurate information about the news they are reading on social media channels; especially important in situations where the news is related to business decisions, such as investment decisions or new product launches. A fake news detection model can help to reduce the risk of businesses being harmed by fake news and increase trust in organizations. For example, a fake news about a company's financial performance can lead to investors selling their shares, which could damage the company's stock price. When investors and customers know that companies are taking steps to combat fake news, they are more likely to believe the information that they are being told. This can help businesses to protect their reputation and avoid negative publicity. So, businesses can use the detector model and can enhance their brand reputation to identify and remove fake news stories that are damaging their brand reputation. This can help businesses to improve the public's perception and make it more attractive to customers and investors. Overall, a misinformation or disinformation detection model development can have a number of positive implications for organizations. By helping to improve decision-making, reduce risk, increase trust, improve public relations, and enhance brand reputation, a fake news detection model can be a valuable tool for businesses in today's digital age.

Feature engineering, which plays an important role in the development of an explainable fake news detection system by shaping the features used in the ML models, is an interconnected concept with explainable AI as it affects the model's interpretability.

In future research work, multimodal visual and text variables identified in this research could be used to develop a single web application or dashboard that could help social media managers and online users to understand the consumption of their own misinformation and disinformation on daily/hourly basis. Our proposed model can also be applied to other broad and diversified applications (e.g. big NLP datasets, biomedical and clinical datasets, etc.) in which the datasets have a large number of features or variables.

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